

VR-Sets.

An Operational Universe of Sets as Revealing Functionalities,
on the Brouwer Path

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Abstract

We reconstruct set theory inside the VR programme as an *operational* universe: a set is a functionality that, on a query, reveals its members — formally, a pointed graph — and the identity of sets is a *witnessed bisimulation*, carried as a relation and never collapsed to a quotient. On this carrier we obtain, machine-checked in Lean 4 and almost entirely below the `Classical.choice` floor: membership, strong extensionality (identity is exactly co-membership), the empty set, pairing, union, the von Neumann naturals and ω , and the Separation and Replacement schemas. Well-foundedness is not an axiom but a *predicate* `IsGrounded`: the ZFC universe and the anti-foundational (Aczel) universe are two fragments of one operational domain, not a foundational choice one is forced to make. Version 1.0.3 corrects the central error of earlier versions. The earlier text claimed the operational universe is *countable* and derived from this that the Axiom of Choice is a free theorem and that paradoxes such as Banach–Tarski cannot arise. That was a Bishop reading and it contradicts the operational continuum, which the VR cycle built on the *Brouwer* path. We replace it: the operational universe — sets *as becoming* — is genuinely *non-enumerable* (a machine-checked, choice-free diagonal), while the *describable* register (sets given by a finite rule) is countable. The Axiom of Choice is therefore *not* an operational theorem; only dependent choice is. The absence of Banach–Tarski is re-explained through conservativity, not through countability.

Keywords: operational set theory, non-well-founded sets, bisimulation, strong extensionality, Brouwer, choice sequences, axiom of choice, constructive foundations, Lean 4.

Contents

1	Introduction and the change in Version 1.0.3	2
2	The carrier: sets as revealing functionalities	3
2.1	Operational sets	3
2.2	Operational identity: witnessed bisimulation, not a quotient	3
2.3	Membership and strong extensionality	3
3	The operations	4
3.1	Infinity	4
3.2	Separation and Replacement	4
4	Foundation is a predicate, not a choice	5
5	Numbers	5

6	Cardinality, the diagonal, and choice — the corrected core	6
6.1	The operational universe is non-enumerable	6
6.2	The describable register is countable	6
6.3	The Axiom of Choice is not a theorem	7
6.4	Why Banach–Tarski does not arise — through conservativity, not counting	7
7	Position and honest boundaries	7
	Acknowledgement of AI assistance	8
	References	8

1 Introduction and the change in Version 1.0.3

VR-Sets realises the set-theoretic layer of the VR programme. A set is not a “thing that contains elements”; it is an *operation* — a functionality that, when queried, *reveals* members, which are themselves functionalities. This reading is constitutive, not decorative: every notion below (membership, identity, the operations, well-foundedness) is defined in its terms, and every claim is checked by the Lean 4 kernel with its axiom cost recorded.

This version revises the foundational core. Earlier versions presented the operational universe through a classical surrogate (mathlib’s well-founded `ZFSet`) and argued, in §VI, that the universe is *countable*, whence the Axiom of Choice holds as a theorem and the classical “pathologies” (non-measurable sets, Banach–Tarski) do not arise. Two things were wrong with that. First, countability is false of the surrogate (`ZFSet` is the full classical uncountable universe), so the claim was carried by prose, not by the machine. Second, and more deeply, the countability reading is a *Bishop* reading, and it is incompatible with the operational continuum that the VR cycle actually constructed, which took the *Brouwer* path: there the space of “becoming” (lawless choice sequences) is non-enumerable, choice-free, and Cantor’s diagonal *works*.

We resolve the tension on the side the cycle already chose. The operational universe of sets is rebuilt *from scratch*, as functionalities with witnessed-bisimulation identity (§2), and we prove (§6) that this universe — sets *as becoming* — is *not* enumerable, by a self-contained diagonal that is choice-free. The countable register is the register of *descriptions*: sets given by a finite rule. The contrast *done countable / becoming non-enumerable* is the corrected §VI. The Axiom of Choice loses its status as a free theorem; this is not a loss but the honest price of a genuinely uncountable operational continuum, and it aligns VR-Sets with VR-Forms (conservativity) and with the operational-continuum work.

What is machine-checked. The development lives in the Lean 4 formalisation of the VR cycle, module `VRCycle.SetsOp` (files `Pointed`, `Builder`, `Closure`, `Omega`, `Extensionality`, `Congruence`, `Becoming`, `Describable`, `Schemas`, `Grounded`). It is self-contained: it imports nothing from mathlib for its core, precisely so that its axiom profile is visible. Throughout, we name the Lean object backing each statement. Almost every object is *axiom-free* or depends only on `propext`; the single exception is flagged honestly in §6.

2 The carrier: sets as revealing functionalities

2.1 Operational sets

Definition 2.1 (Operational set; `OpSet`). An *operational set* is a pointed graph

$$x = (V, E, pt), \quad V : \text{Type}, \quad E : V \rightarrow V \rightarrow \text{Prop}, \quad pt : V.$$

Read $E a b$ as “querying the functionality at vertex b reveals a as a member”; pt is the set itself. The member of x sitting at a vertex a is the same graph re-pointed there, $x.\text{child } a := (V, E, a)$.

This is exactly the language in which Aczel’s anti-foundation axiom is stated (decoration of a graph), but we take *no quotient*: the identity of sets is supplied operationally, below.

2.2 Operational identity: witnessed bisimulation, not a quotient

Definition 2.2 (Bisimulation; operational identity $\approx$). A relation $R : x.V \rightarrow y.V \rightarrow \text{Prop}$ is a *bisimulation* between x and y if related vertices reveal matching members in both directions: for all a, b with $R a b$,

$$(\forall a', x.E a' a \Rightarrow \exists b', y.E b' b \wedge R a' b') \wedge (\forall b', y.E b' b \Rightarrow \exists a', x.E a' a \wedge R a' b').$$

We write $x \approx y$ (x and y are *operationally identical*) when some bisimulation relates their points: $\exists R, R x.pt y.pt \wedge R$ is a bisimulation.

Proposition 2.3 ($\approx$ is an equivalence; `Equiv.refl/symm/trans`). $\approx$ is *reflexive, symmetric and transitive*. Each is axiom-free in *Lean* (no `propext`, no `Quot.sound`, no `Classical.choice`).

The methodological point is decisive for the VR reading. To turn “bisimilar” into a *type-level equality* — to form the quotient — one must Skolemise the existential matches of Definition 2.2 into functions; that step is exactly VR’s transit asymmetry (operational $\rightarrow$ formal is free, formal $\rightarrow$ operational has no mechanism), and it is precisely where `mathlib`’s coinductive set construction invokes `Classical.choice`. We therefore do *not* quotient. Identity remains a relation carried with its witness: a *performed* bisimulation, not a given completion. This is “doing, not being” made into the construction, and it is why the whole development stays below the choice floor.

2.3 Membership and strong extensionality

Definition 2.4 (Membership; `Mem`). $z \in x$ iff some member-vertex of x is operationally identical to z : $\exists a, x.E a x.pt \wedge z \approx x.\text{child } a$.

Theorem 2.5 (Membership respects identity; `mem_congr`). If $y \approx y'$ then $z \in y \Rightarrow z \in y'$. (Axiom-free.)

Theorem 2.6 (Strong extensionality; `ext`, `equiv_iff_same_mem`). Two operational sets are identical iff they have the same members:

$$x \approx y \iff \forall z, (z \in x \leftrightarrow z \in y).$$

Both directions are axiom-free.

Theorem 2.6 is the heart of the carrier. The forward direction is Theorem 2.5 applied both ways; the backward direction is the coinductive content — one *builds* a bisimulation out of the member-matching (relate the two roots; relate any further pair whose rooted subtrees are already identical; the root step uses the hypothesis, the subtree step unfolds and re-wraps the witnessing bisimulation). It is the AFA-style *strong* extensionality, but kept choice-free precisely because identity was never quotiented. It is what licenses $\approx$ as a faithful equality of sets.

3 The operations

All operations are instances of one constructor.

Definition 3.1 (Family supremum; `sup`). For a family $c : I \rightarrow \text{OpSet}$, the set $\text{sup } c$ whose members are exactly the ci is the pointed graph with a fresh root revealing each component's point, components kept disjoint.

Theorem 3.2 (`mem_sup`). $z \in \text{sup } c \iff \exists i, z \approx ci$. (*Axiom-free.*)

From this single law the standard operations and their membership conditions follow; each is axiom-free or depends only on `propext`.

- **Empty set** $\emptyset$ (`emptySup`): the empty family. $z \notin \emptyset$ (`not_mem_emptySup`). $\emptyset$ is the nullary operation — the functionality that reveals nothing — matching the base of the VR formal system.
- **Singleton** $\{a\}$ (`singleton`): $z \in \{a\} \iff z \approx a$.
- **Unordered pair** $\{a, b\}$ (`pair`): $z \in \{a, b\} \iff z \approx a \vee z \approx b$.
- **Union** $\bigcup x$ (`union`): $z \in \bigcup x \iff \exists y, y \in x \wedge z \in y$ (`mem_union`).
- **Binary union** $a \cup b$ and **successor** $\text{succ } a := a \cup \{a\}$: $z \in \text{succ } a \iff z \in a \vee z \approx a$ (`mem_succ`).

Proposition 3.3 (Operations respect identity; `*_congr`). `singleton`, `pair`, `∪` and `succ` all respect $\approx$ (proved through Theorem 2.6).

3.1 Infinity

Definition 3.4 (Von Neumann naturals and ω ; `vn`, `omega`). $\text{vn } 0 := \emptyset$, $\text{vn}(n+1) := \text{succ}(\text{vn } n)$; $\omega := \text{sup } \text{vn}$.

Theorem 3.5 (Infinity; `mem_omega`, `empty_mem_omega`, `omega_succ_closed`). The members of ω are exactly the $\text{vn } n$ up to $\approx$; $\emptyset \in \omega$; and ω is closed under successor for any member: $x \in \omega \Rightarrow \text{succ } x \in \omega$. (*Axiom-free, resp. propext.*)

3.2 Separation and Replacement

A genuine property or function *of sets* cannot depend on the particular graph presenting a set; it must respect $\approx$. With that — and no decidability assumption — the two schemas hold.

Theorem 3.6 (Separation; `sep`, `mem_sep`). For a species p respecting $\approx$, the set $\{z \in x \mid pz\}$ exists and $z \in \{z \in x \mid pz\} \iff z \in x \wedge pz$. (*propext.*)

Theorem 3.7 (Replacement; `repl`, `mem_repl`). For a rule F respecting $\approx$, the image $\{Fz \mid z \in x\}$ exists and $w \in \{Fz \mid z \in x\} \iff \exists z, z \in x \wedge w \approx Fz$. (*propext.*)

4 Foundation is a predicate, not a choice

Classical set theory must *decide*, at the level of its axioms, whether $\in$ -descent always terminates: ZFC posts Foundation, Aczel’s theory posts the Anti-Foundation Axiom (AFA). These are two incompatible commitments about nested membership, and a classical theory is forced to pick one.

VR is not. On the operational carrier, well-foundedness is simply a *property of a set’s graph*, observed and not postulated.

Definition 4.1 (Groundedness; `isGrounded`). x is *grounded* iff its point is accessible under the reveal relation: $\text{Acc}(E)x.pt$ — every membership descent from x terminates.

Theorem 4.2 (Groundedness is a property of the set; `isGrounded_iff`). *If $x \approx y$ then x grounded $\iff y$ grounded. (Axiom-free.)*

Theorem 4.3 (The ZFC fragment is hereditary; `mem_grounded`). *A member of a grounded set is grounded. (Axiom-free.)*

Theorem 4.2 is what makes “well-founded” a well-defined predicate on operational sets (it is transported across a bisimulation; the key lemma `acc_bisim` carries accessibility along R). Hence:

- the **ZFC fragment** is $\{x \mid x \text{ grounded}\}$ — $\emptyset$ is grounded (`empty_isGrounded`); the fragment is closed downward under $\in$ (Theorem 4.3);
- the **ZFA fragment** is the whole universe. The Quine atom $A = \{A\}$ is a perfectly good operational set — one vertex with a self-loop (`quine`), satisfying $A \in A$ (`quine_self_mem`) — and it is *not* grounded (`quine_not_grounded`). No anti-foundation axiom is invoked; the graph simply has a cycle.

VR thus holds ZFC and ZFA *natively*, as fragments of one universe distinguished by a predicate, and faces neither Foundation nor AFA as a postulate. Identity is uniform throughout (witnessed bisimulation), which is exactly the AFA notion of equality, applied to grounded and non-grounded sets alike. The grounded fragment additionally enjoys $\in$ -acyclicity: no von Neumann natural is a member of itself (`vn_not_self_mem`, by strong induction).

5 Numbers

The natural numbers are the von Neumann naturals of §3, and they are genuinely distinct as operational sets.

Theorem 5.1 (The operational naturals are distinct; `vn_inj`). $vn\ n \approx vn\ m \Rightarrow n = m$. (*propext.*)

The proof is the classical one read operationally: the members of $vn\ m$ are exactly $vn\ 0, \dots, vn(m-1)$ (`vn_mem_iff`), no $vn\ n$ is a member of itself (`vn_not_self_mem`), and these force injectivity by the trichotomy of $\mathbb{N}$.

Remark 5.2 (Two reals — stated honestly). The integers, rationals, reals and complex numbers of the VR cycle come in two distinct guises, and they must not be conflated. *VR-Numbers* builds $\mathbb{Z}_{VR}, \mathbb{Q}_{VR}, \mathbb{R}_{VR}, \mathbb{C}_{VR}$ proved isomorphic to the corresponding classical (`mathlib`) fields; that $\mathbb{R}_{VR} \cong \mathbb{R}$ is the *full, uncountable* classical line. The operational continuum of the VR cycle is a *different* object: the becoming space of choice sequences and the Bishop reals over $\mathbb{Z}$, built below the choice floor and *not* claimed isomorphic to the classical line. The operational reals are the becoming side; $\mathbb{R}_{VR}$ is the classical correlate. They coexist; their precise relation is the open Bishop/Brouwer question of the programme.

6 Cardinality, the diagonal, and choice — the corrected core

This section replaces the central claims of the earlier §VI.

6.1 The operational universe is non-enumerable

Theorem 6.1 (Non-enumerability of the universe; `universe_not_enumerable`). *No $\mathbb{N}$ -indexed enumeration of operational sets is surjective up to operational identity:*

$$\neg \exists e : \mathbb{N} \rightarrow \text{OpSet}, \forall x, \exists n, e\ n \approx x.$$

*The proof is choice-free (**propext** only).*

The proof is a self-contained Cantor diagonal carried out *inside* the operational universe. Given a candidate enumeration e , form the diagonal set

$$D := \{ \text{vn } n \mid \text{vn } n \notin e\ n \},$$

a legitimate operational set (a **sup** over the subtype of indices n with $\text{vn } n \notin e\ n$; no decidability is needed). By the distinctness of the naturals (Theorem 5.1), $\text{vn } k \in D \iff \text{vn } k \notin e\ k$. If e were surjective, some $e\ m \approx D$, whence by extensionality $\text{vn } m \in e\ m \iff \text{vn } m \in D \iff \text{vn } m \notin e\ m$ — a contradiction.

This is the Brouwerian content of VR-Sets. The diagonal does *not* fail and the universe is *not* countable. Its meaning is the cut between the *done* and the *becoming*: the countable register of what has been performed never exhausts the universe of sets as becoming. (In the operational-continuum work the same phenomenon appears for the power set of ω : the characteristic functions $\mathbb{N} \rightarrow \text{Bool}$ are non-enumerable, choice-free.)

6.2 The describable register is countable

Definition 6.2 (Finite descriptions; `Desc`, `IsDescribable`). Let `Desc` be the finite inductive syntax with constructors `∅`, `pair`, `union`, `ω`, and let `eval` : `Desc` → `OpSet` be its interpretation. A set x is *describable* if $x \approx \text{eval } d$ for some $d : \text{Desc}$.

Theorem 6.3 (The describable register is countable; `describable_countable`). *There is an $\mathbb{N}$ -indexed enumeration reaching every describable set (up to $\approx$); the enumeration is exhibited as the rule $n \mapsto \text{eval}(\text{decode } n)$.*

Remark 6.4 (Honest axiom note). Theorem 6.3 is the only object in the development that depends on `Classical.choice`; the dependence is *borrowed*, not intrinsic. It enters solely through `mathlib`’s proof that the natural-number pairing is invertible (`Nat.unpair_pair` / `Nat.pair_eq_pair` both use choice; the pairing `Nat.pair` itself is axiom-free). The fact is constructively true; `mathlib` merely proves it with incidental choice. We keep the theorem (it is a genuine enumeration) and record the cost. The countability of a register of *descriptions* is in any case a statement of the formal register — meta-accounting over a finite syntax — where, in the VR apparatus, classical reasoning is admitted with its cost made visible.

Theorems 6.1 and 6.3 together *are* the corrected §VI. The earlier claim “the operational universe is countable, hence AC is free” is false; the correct statement is the contrast: **the describable (done) register is countable, the universe (becoming) is not.**

6.3 The Axiom of Choice is not a theorem

The earlier §VI argued that AC holds in VR-Sets because the universe is countable, so every family can be enumerated and “the first element of each” chosen. That argument is unsound here: its premise (every family has a countable enumeration) is exactly what Theorem 6.1 refutes. “AC is free” and “the continuum is genuinely uncountable” are two settings of one switch, and the VR cycle has set it on the Brouwer side.

The honest operational picture is therefore:

- **Dependent choice** is operational and choice-free: a history-dependent rule determines a branch by recursion (the `operational_dependent_choice` of the operational-continuum work).
- **Weak continuity** (WC-N) holds operationally for finite-information functionals (machine-checked there), and is classically false — the cleanest two-register exhibit.
- **Full AC** — a selection over an uncountable family with no rule — has no operational correlate. It is a label of the formal register, not an operational theorem.

6.4 Why Banach–Tarski does not arise — through conservativity, not counting

The earlier text located the absence of Banach–Tarski in the countability of the universe (no uncountable sphere, no non-measurable Vitali set). With countability gone, the explanation must change — and the correct one is sharper and ties VR-Sets to VR-Forms.

The Banach–Tarski construction is a *form applied to a form* ($T \rightarrow T$): it builds, in the formal register, a completed uncountable totality (the sphere as a set of points) together with a selection by *full* AC (the Vitali set). Neither ingredient has an operational correlate: the completed uncountable totality is a formal-register referent, and full AC, as above, is a formal-register label. By the invariance of the operational under the application of form — the ground of VR-Forms’ conservativity result, which is itself machine-checked — a formal-register construction *does not transit to the operational register*: form has no power to manufacture an operational object where none was built. The paradox therefore lives entirely in the formal register and never returns through the operational floor. Its absence is a *conservativity* phenomenon, not a *cardinality* accident.

7 Position and honest boundaries

What is established. The operational universe of sets is built without classical scaffolding: a set is a revealing functionality, identity is a witnessed bisimulation (no quotient), and on this carrier membership, strong extensionality, $\emptyset$, pairing, union, ω , Separation and Replacement, the distinctness of the naturals, and the predicate character of well-foundedness are all machine-checked — almost entirely axiom-free or `propext`-only. The corrected cardinality core (universe non-enumerable; describable register countable) is machine-checked, the substantive half choice-free.

What is open or deferred.

- **Bishop vs Brouwer.** The operational reals (becoming) and the classical $\mathbb{R}_{VR}$ coexist; which is canonical for the VR continuum is the programme’s open question. This version commits the *set* universe to the Brouwer reading, consistently with the operational-continuum work.

- **The metalanguage.** Lean treats the underlying types classically; the operational reading is carried in the metalanguage and tracked by axiom profile. “Below the choice floor” is a statement about the construction’s audited dependencies, exhibited build-time, not a claim that the ambient logic is intuitionistic.
- **The borrowed choice.** The countability of the describable register inherits one `Classical.choice` from `mathlib`’s pairing inversion (Remark 6.4); removing it would mean re-deriving the arithmetic choice-free, which is orthogonal to the foundations and left aside.
- **Power set.** There is no operational “power-set operation” returning a completed set; the power set of ω is the becoming, and the non-enumerability of §6 is precisely the operational content of “ $\wp(\omega)$ ”. CH is a statement about a formal-register referent; it is read philosophically, not coded.

Relation to the earlier version. Versions through 1.0.2 stand as the *classical-surrogate* presentation (the ZFC axioms realised over `ZFSet`); their §VI cardinality claims are superseded by §6 here. The mathematics that was genuinely machine-checked there (the ZFC axioms as closure facts over the surrogate) is unaffected; what is corrected is the operational reading of cardinality and choice.

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References

- P. Aczel, *Non-Well-Founded Sets*, CSLI Lecture Notes 14, 1988.
- E. Bishop, *Foundations of Constructive Analysis*, McGraw–Hill, 1967.
- L. E. J. Brouwer, collected works on choice sequences and the continuum.
- J. von Neumann, *Zur Einführung der transfiniten Zahlen*, 1923.
- V. Reznik, *VR. A Formal System* (DOI 10.5281/zenodo.20324240); *VR-Numbers*; *VR-Forms*; *VR-Sets-ZFA*. Zenodo, 2026.
- The VR Cycle Lean 4 formalisation, <https://github.com/inventor1975/VRCycle>, module `VRCycle.SetsOp`.